User Guide

Running the coupled SWAT$^+$/MODFLOW 6 PFAS fate-and-transport model

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This guide explains, step by step and without assuming a programming background, *how to run* the coupled surface-water/groundwater PFAS model described in the main paper. For the scientific basis, parameter sources, and validation, see the paper and its Supplementary Material; for how to obtain, configure, and run the model, everything is here.

There are two ways to get a working model: **(A)** generate it on the SWATGenX website (no installation; best if you are in the United States), or **(B)** build and run it on your own Windows desktop (best to change the model, run scenarios, or work offline). Pick one.

Contents

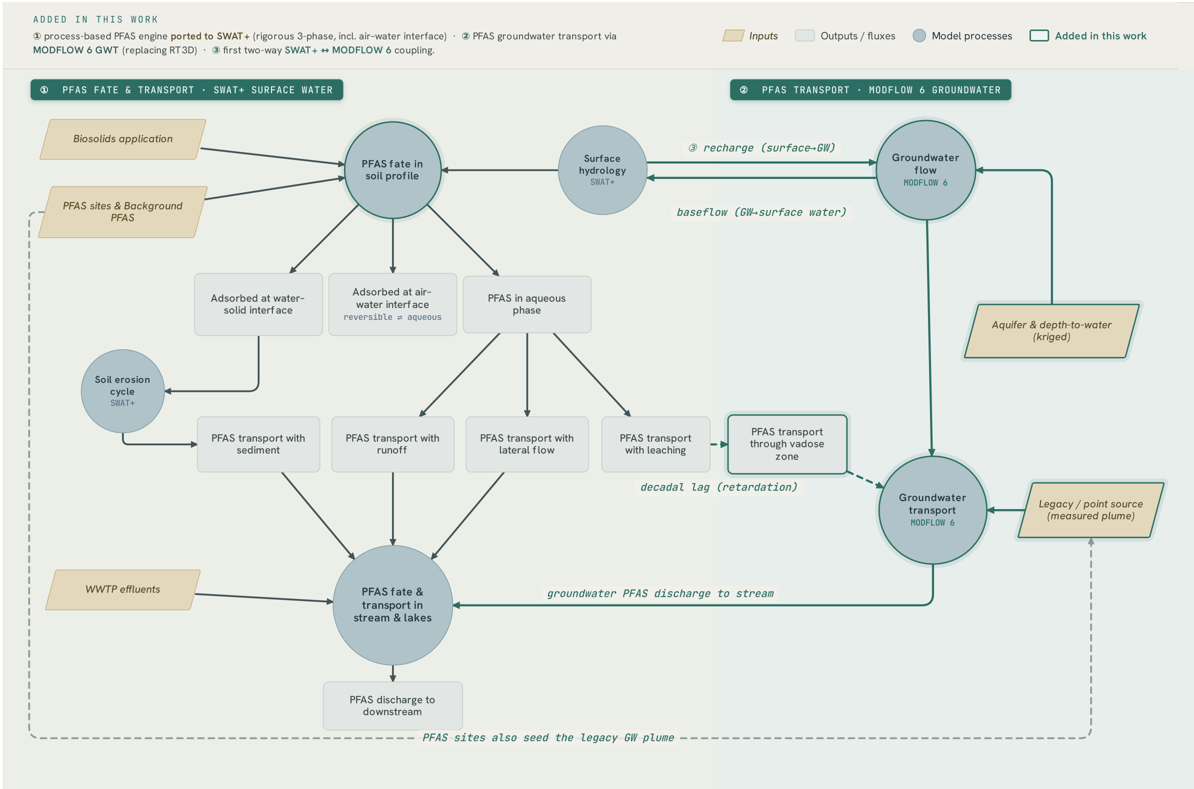

Figure G1: Conceptual model. PFAS is partitioned in the soil profile across the aqueous, solid, and air–water-interface phases, transported to streams with runoff, lateral flow, and sediment (existing SWAT$^+$ surface leg), and — new in this work — carried to groundwater and back: the daily two-way SWAT$^+$↔MODFLOW 6 water exchange (recharge down, baseflow up), the (slow, retardation-lagged) vadose pathway, the legacy groundwater plume, and the groundwater PFAS discharge to streams.

1 The easy way: build it on swatgenx.com

1. Go to `https://swatgenx.com` and create a free account (Figure A1).
2. Find your watershed by its USGS streamgage number or by clicking the stream network (Figure A2).
3. Choose what to build (Figure A3):
   - **SWAT$^+$ (surface water)** — available for the entire continental United States.
   - **SWAT$^+$ + MODFLOW 6 (surface water + groundwater)** — currently available for the **Michigan Lower Peninsula** only; **more states are coming soon**.
   - Turn on the **PFAS** option for PFAS fate and transport.
4. Click **Build**. When it finishes you get a download link and an online results dashboard (Figure A4).

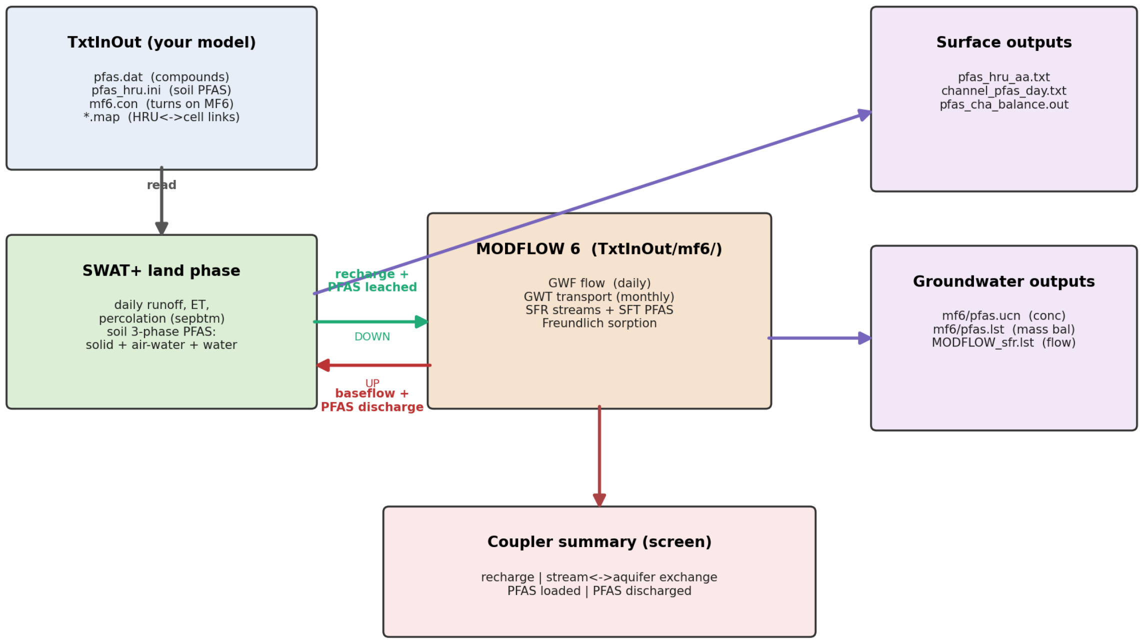

Figure G2: How data flows through a coupled run. The engine reads the `TxtInOut` files, runs the SWAT$^+$ land phase, sends recharge and leached PFAS *down* to MODFLOW 6, returns baseflow and discharged PFAS *up* into the streams, and writes the surface, groundwater, and exchange outputs.

5. Download the model folder. Inside it is a folder named `TxtInOut` — that *is* the model. You can use the website results, or run it yourself with the engine (Section 2.5).

If MODFLOW 6 is not yet available for your state, you can still build SWAT$^+$ everywhere and add groundwater later.

2 The full way: run it on your Windows desktop

2.1 Get the code

Install *Git for Windows* (`https://git-scm.com`; accept the defaults), open *Git Bash*, and copy the code:

```
git clone https://github.com/rafiei-vahid/swatplus.git
cd swatplus
git checkout feat/pfas-surface-water
```

This gives you the SWAT$^+$ engine with the PFAS and MODFLOW 6 coupling built in.

2.2 Get an example model

Download the example Rogue River PFAS model from the paper's data page and unzip it. You will get a `TxtInOut` folder of text files — that is the model.

2.3 Compile the engine

You need the *Intel Fortran compiler* (`ifx`), free in the Intel oneAPI HPC Toolkit (`https://www.intel.com/oneapi`). Open the "Intel oneAPI command prompt" and build (Figure B1):

```
cd swatplus
cmake -B build -S . -DCMAKE_Fortran_COMPILER=ifx
cmake --build build
```

You now have `swatplus.exe` in `build`. Copy it into your `TxtInOut` folder. (For surface-water SWAT$^+$ only, that is all; the groundwater coupling switches on only when the control file below is present.)

2.4 Tell the engine to use MODFLOW 6

The engine runs MODFLOW 6 *only* if it finds a small text file named `mf6.con` in `TxtInOut`. No file = plain SWAT$^+$ (nothing changes); with the file = live daily two-way coupling.

The MODFLOW 6 model lives in a **sub-folder of** `TxtInOut` — by convention `TxtInOut/mf6/` — containing the MODFLOW 6 files (`mfsim.nam`, `.dis`, `.npf`, `.sfr`, ...). The engine loads the MODFLOW 6 library (`libmf6`) and drives it from there. Copy this `mf6.con` template (text after `!` is ignored):

```
./mf6      ! 1: folder (inside TxtInOut) holding mfsim.nam
1          ! 2: groundwater FLOW step, in days (1 = daily)
30         ! 3: groundwater TRANSPORT step, in days (30 = monthly)
1.0        ! 4: (optional) recharge multiplier for calibration
           ! 5: (optional) full path to libmf6 if not on the search path
```

Three small map files also go in `TxtInOut` (already present in the website and example models; you build them only for a brand-new watershed):

File	What it does
`mf6_recharge.map`	links each SWAT$^+$ field (HRU) to the MODFLOW cells under it, so percolation becomes recharge
`mf6_baseflow.map`	links each MODFLOW stream reach to its SWAT$^+$ channel, so groundwater discharge returns to the right stream
`pfas_leach.map`	(PFAS only) links fields to cells so soil-leached PFAS becomes a groundwater source

2.5 Run it

From the Intel oneAPI command prompt, inside `TxtInOut`, run `swatplus.exe`. You will see day-by-day progress, a line `MF6 COUPLER: active` if `mf6.con` is present, and a summary of the water and PFAS exchanged between surface water and groundwater (Section 5).

3 Turning PFAS on

PFAS is controlled by two files in `TxtInOut` (already present in the example and website models): `pfas.dat` (the compounds and their chemistry; no file = no PFAS) and `pfas_hru.ini` (the starting soil PFAS in each field, plus its sorption numbers).
`pfas.dat` — one row per compound:

```
 id  name      mw        sol      kl       lm       percop
  1  PFOS    0.50013   680.0    0.137   2500.0    0.20
  0  END     0.0       0.0      0.0      0.0       0.0
```

`pfas_hru.ini` — starting soil PFAS and sorption, per field:

```
hru 1 nly 4                                   <- field 1 has 4 soil layers
1                                             <- number of PFAS in this field
0.0400 0.0600 0.0600 0.0500                   <- grain size d50 (mm), per layer
1 1.536e+07 5.610e+06 6.591e+06 1.107e+07     <- id + starting soil PFAS per layer
1 450.2 227.8 227.8 365.7                     <- Freundlich kf per layer
1 0.4300 0.3800 0.3800 0.4500                 <- Freundlich n per layer
1 1.0                                         <- sediment enrichment ratio
```

4 The PFAS parameters added to SWAT$^+$

Table G1 lists every input the PFAS module adds, in the spirit of the parameter list in Rafiei et al. (2023, *Water Research*). The soil three-phase model (aqueous / solid / air–water interface) uses them all. The air–water interfacial area is computed, not entered: $A_{aw} = 6(1-\phi)(1-S_w)/d_{50}$, with ϕ porosity and S_w saturation — so `d50` and soil moisture matter (drier, finer soils hold more PFAS at air–water interfaces).

5 Reading the output

After a run, these files appear in `TxtInOut` (surface) and `TxtInOut/mf6/` (groundwater).
The engine also prints an end-of-run **coupling summary**: the recharge delivered to groundwater, the stream–aquifer exchange (*gaining* = groundwater feeding streams, *losing* = streams leaking down), the PFAS loaded to the aquifer, and the PFAS discharged back to the streams. Those last two numbers *are* the surface-water/groundwater PFAS exchange.
A real run ends:

Table G1: PFAS parameters added to SWAT$^+$.

Parameter	File	Units	Meaning
`mw`	pfas.dat	$\mathrm{kg\,mol^{-1}}$	molecular weight
`sol`	pfas.dat	$\mathrm{mg\,L^{-1}}$	maximum aqueous solubility
`kl`	pfas.dat	$\mathrm{L\,nmol^{-1}}$	Langmuir constant, air–water interface (K_L)
`lm`	pfas.dat	$\mathrm{nmol\,m^{-2}}$	Langmuir maximum surface coverage (Γ_{max})
`percop`	pfas.dat	– (0–1)	split of PFAS between percolation and runoff
`sol_pfas`	pfas_hru.ini	$\mathrm{kg\,ha^{-1}}$	starting PFAS mass per soil layer
`kf`	pfas_hru.ini	$(\mathrm{nmol\,kg^{-1}})/(\mathrm{nM})^n$	Freundlich solid-phase sorption coefficient
`nf` (`n`)	pfas_hru.ini	–	Freundlich exponent (sorption non-linearity)
`d50`	pfas_hru.ini	mm	median grain diameter (sets air–water interfacial area)
`enr`	pfas_hru.ini	–	enrichment ratio for PFAS on eroded sediment

File	What is in it
`pfas_hru_aa.txt`	per-field PFAS budget: soil PFAS start/end and where it went — surface runoff, lateral flow, *percolation toward groundwater*, sediment, and what remained
`channel_pfas_day.txt`	daily in-stream PFAS concentration and load per channel — the signal compared to monitoring data
`pfas_cha_balance.out`	in-stream PFAS mass balance (in vs out vs stored)
`mf6/pfas.ucn`	groundwater PFAS concentration in every aquifer cell, over time
`mf6/pfas.lst`	groundwater transport mass budget, incl. the `PERCENT DISCREPANCY` (your mass-balance check; should be near zero)
`mf6/MODFLOW_sfr.lst`	groundwater flow budget, incl. the stream–aquifer exchange

```
MF6 COUPLER summary over 91 coupled days:
  recharge (sum of daily basin depths) =   5.2015E+01 m
  SFR exchange: gaining   2.9579E+07  losing  -2.6348E+07 m3
  PFAS discharged to stream =   1.4780E+00 kg
Execution successfully completed
```

and `pfas_hru_aa.txt` looks like:

```
hru pfas  init_kgha   final_kgha  surq_kgha   ... perc_kgha   ...
  1    1  4.24941E-03 4.24941E-03 1.53459E-09 ... 0.00000E+00 ...
```

Here `perc_kgha`= 0 means this compound did not leach to groundwater over the run — a real result for strongly-sorbing PFOS on a short run; mobile compounds such as PFHxA show non-zero leaching.

6 File schemas: everything we add to `TxtInOut`

Beyond the standard SWAT$^+$ and MODFLOW 6 files, the PFAS and coupling modules add the files below. All are plain text, whitespace-delimited, free-format (column spacing is not significant). This section is the exact format of each, grouped as *input*, *coupler*, and *output*.

6.1 Input files (you provide these)

`pfas.dat` — compound database; one row per compound, ended by a row with `id` = 0 (`END`). First two lines are headers.

Field	Type	Units	Meaning
`id`	int	–	compound number (1, 2, . . .; 0 ends the list)
`name`	text(16)	–	compound name (e.g. PFOS)
`mw`	real	$\mathrm{kg\,mol^{-1}}$	molecular weight
`sol`	real	$\mathrm{mg\,L^{-1}}$	maximum aqueous solubility
`kl`	real	$\mathrm{L\,nmol^{-1}}$	Langmuir K_L (air–water interface)
`lm`	real	$\mathrm{nmol\,m^{-2}}$	Langmuir Γ_{max} (air–water interface)
`percop`	real	– (0–1)	percolation/runoff split of mobile PFAS

`pfas_hru.ini` — starting soil PFAS and sorption, one block per HRU (after two header lines), in HRU order:

```
hru <id> nly <nlayers>            header for this HRU
<num_pconta>                      number of PFAS compounds in this HRU
<d50_L1> <d50_L2> ...             median grain diameter per layer (mm)
<pid> <solpfas_L1> <solpfas_L2>.. starting soil PFAS per layer (kg/ha) [1 line/PFAS]
<pid> <kf_L1> <kf_L2> ...         Freundlich kf per layer            [1 line/PFAS]
<pid> <nf_L1> <nf_L2> ...         Freundlich n  per layer            [1 line/PFAS]
<pid> <enr>                       sediment enrichment ratio          [1 line/PFAS]
```

`pfas_calib.dat` (optional) — one line, two global multipliers; missing ⇒ both 1.0:

```
<soil_scale> <koc_scale>   e.g.  0.11 1.0
```

`soil_scale` scales the initial soil PFAS pool; `koc_scale` scales the in-stream sorption.

`pfas_cha.dat` (optional) — per-compound in-stream transport parameters; missing ⇒ built-in defaults. Columns: `id`, `name`(16), `koc` ($\mathrm{m^3\,g^{-1}}$), `settle` ($\mathrm{m\,d^{-1}}$), `resus` ($\mathrm{m\,d^{-1}}$), `bury` ($\mathrm{m\,d^{-1}}$), `act_dep` (m).

6.2 Coupler files (link SWAT$^+$ and MODFLOW 6)

`mf6.con` — the switch that turns on MODFLOW 6 (Section 2.4). One value per line; text after `!` ignored.

Line	Required	Meaning
1	yes	MODFLOW 6 workspace folder, relative to `TxtInOut` (e.g. `./mf6`)
2	yes	groundwater FLOW step, days (1 = daily)
3	yes	groundwater TRANSPORT step, days (30 = monthly)
4	no	recharge multiplier (default 1.0)
5	no	full path to `libmf6` (default: system search path)

The three map files link the two grids. The first line is a header (counts); each following line is one link.

File	Header line	Each row
`mf6_recharge.map`	`N_ENTRIES N_CELLS NCOL N_HRU`	`cell_index hru weight`
`mf6_baseflow.map`	`N_LINKS N_REACHES`	`reach_index gis_channel`
`pfas_leach.map`	`N_ENTRIES N_SRCCELLS`	`src_index hru overlap_ha`

- `cell_index` = `row`×`NCOL`+`col` (0-based, row-major, into MODFLOW's recharge array); `weight` = (HRU–cell overlap area) / (cell area).
- `reach_index` is the 0-based MODFLOW SFR reach; `gis_channel` is the SWAT$^+$ channel GIS id it discharges to.
- `src_index` is the 0-based entry in the groundwater PFAS source list; `overlap_ha` is the HRU–cell overlap area (ha).

6.3 Output files (the model writes these)

`pfas_hru_aa.txt` — run-average per-HRU soil-PFAS budget. Columns: `hru`, `pfas`, `init_kgha`, `final_kgha`, `surq_kgha` (surface runoff), `latq_kgha` (lateral flow), `perc_kgha` (leached toward groundwater), `sed_kgha` (on sediment), `resid_kgha` (left in soil); all kg ha^{-1}.

`pfas_balance.out` — basin land-phase PFAS mass balance (kg over the run) per compound: `init_kg`, `final_kg`, `surq_kg`, `latq_kg`, `perc_kg`, `sed_kg`, plus the maximum per-HRU closure residual (a numerical check).

`channel_pfas_day.txt` — daily in-stream PFAS per channel. Columns: `jday`, `mon`, `day`, `yr`, `unit`, `gis_id`, `name`, `pfas`, then the daily mass terms (kg) `tot_in`, `sol_out`, `sor_out`, `settle`, `resus`, `diffuse`, `bury`, `water`, `benthic`, and the reach outflow concentration `conc_ngL` (ng L^{-1}).

`pfas_cha_balance.out` — in-stream PFAS mass balance (run- cumulative, kg): reach-days with inflow, point-source input, cumulative reach inflow and outflow, and burial (the benthic sink).

The MODFLOW 6 side (`mf6/pfas.ucn`, `mf6/pfas.lst`, `mf6/MODFLOW_sfr.lst`) uses *standard* MODFLOW 6 file formats and is not re-documented here.

7 Troubleshooting

- **No MODFLOW output?** `mf6.con` is missing/misspelled or not in `TxtInOut`; the engine silently runs plain SWAT$^+$.
- **No PFAS numbers?** `pfas.dat` is missing or has no compound rows before `END`.
- **Stops while reading a MODFLOW file?** A path in `mf6.con` is wrong, or `mf6/` lacks a valid `mfsim.nam`.
- **Large `PERCENT DISCREPANCY` in `pfas.lst`?** The transport step (line 3 of `mf6.con`) is too long; reduce it.